

SYDNEY TECHNICAL HIGH SCHOOL

YEAR 12 ASSESSMENT TASK 3

JUNE 2002

MATHEMATICS

EXTENSION 1

Time Allowed: 70 minutes

Instructions:

- * Attempt all questions
- * Answers to be written on the paper provided.
- * Start each question on a new page.
- * All necessary working should be shown.
- * Marks may not be awarded for careless or badly arranged working.
- * This question paper must be stapled on top of your answers.
- * Marks shown are for guidance and may be changed slightly if needed.
- * Standard integrals are attached and may be removed for your convenience.

Name: _____ Teacher: _____

Question 1	Question 2	Question 3	Question 4	Question 5	Question 6	Total

Question 1

- a) Differentiate $y = \cos^{-1} 2x$ 1
- b) Find $\frac{d}{dx}(2^x)$ 1
- c) Find as an exact value $\sin^{-1} \frac{1}{2} + \cos^{-1} \left(-\frac{\sqrt{3}}{2}\right)$ 2
- d) Solve the equation $\ln(x+7) = 2 \ln(x+1)$ 3
- e) (i) Sketch, without the use of calculus, the polynomial $P(x) = (x-1)^2(x+1)^3$ showing the x and y intercepts. 3
- (ii) Hence solve the inequation $P(x) \geq 0$

Question 2

- a) Consider the function $f(x) = e^{x+2}$ 4
- (i) Find the inverse function $f^{-1}(x)$
- (ii) Sketch the graphs of $f(x)$ and $f^{-1}(x)$ on the same number plane. Clearly label each graph and show the intercepts.
- b) The polynomial $P(x) = x^3 + 2x^2 + ax + b$ has a factor of $(x+2)$ and when divided by $(x-2)$ there is a remainder of 12. Find the values of a and b . 4

Question 3

- a) Find $\frac{d}{dx} \log_e(\sin^{-1} x)$ 2
- b) Find $\int_0^{\frac{2}{5}} \frac{dx}{\sqrt{16-25x^2}}$ as an exact value 3
- c) Use the substitution $u = e^x$ to find the exact value of 4

$$\int_0^{\ln \sqrt{3}} \frac{e^x}{1+e^{2x}} dx$$

Question 4

Consider the function $y = \log_e \left(\frac{2x}{2+x} \right)$ where $x < -2, x > 0$

- a) Find the value of x for which $y = 0$. 1
- b) Show that $\frac{dy}{dx} = \frac{2}{x(2+x)}$ and hence state why the 2
function is increasing for all x in the given domain.
- c) Are there any points of inflexion? Justify your answer. 2
(You may use $\frac{d^2y}{dx^2} = \frac{1}{(2+x)^2} - \frac{1}{x^2}$)
- d) Determine the equation of the horizontal asymptote. 1
- e) Sketch the graph of the function showing the features from (a) to 2
(d) above.

Question 5

- a) (i) Find $\frac{d}{dx} (x \tan^{-1} x)$ 4
- (ii) Hence find the exact value of $\int_0^1 \tan^{-1} x \, dx$
- b) Two of the zeros of the cubic polynomial $P(x) = 3x^3 - bx^2 - 27x + 9$ 4
are reciprocals of each other, and two of the zeros of $P(x)$ are opposite in
sign *but equal in magnitude*.
- (i) Find the value of b .
- (ii) Factorise $P(x)$ completely.

Question 6

Consider the function $f(x) = \sin^{-1}(x-1)$

- | | | |
|-------|---|----------|
| (i) | Evaluate $f(0)$ | 1 |
| (ii) | State the domain and range of $y = f(x)$ | 2 |
| (iii) | Draw the graph of $y = f(x)$ | 1 |
| (iv) | The area bounded by the curve $y = f(x)$, the y axis | 4 |

and the line $y = \frac{\pi}{2}$ is rotated about the y axis.

Find the volume of the solid formed.

Question 1 (10 marks)

a) $\frac{dy}{dx} = \frac{-2}{\sqrt{1-4x^2}}$ ①

b) $\ln 2 \cdot 2^x$ ①

c) $\sin^{-1} \frac{1}{2} + \cos^{-1} \left(-\frac{\sqrt{3}}{2}\right)$
 $= \frac{\pi}{6} + \pi - \cos^{-1} \frac{\sqrt{3}}{2}$
 $= \frac{\pi}{6} + \pi - \frac{\pi}{6}$
 $= \pi$ ②

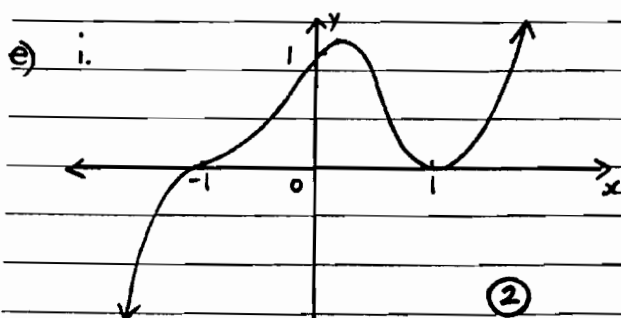
d) $\ln(x+7) = 2 \ln(x+1)$
 $\ln(x+7) = \ln(x+1)^2$
 $x+7 = x^2+2x+1$ ①

$0 = x^2+x-6$

$0 = (x+3)(x-2)$

$x = 2 \text{ or } -3$ ①

$\therefore \underline{x = 2}$ only ($x = -3$ gives $\ln(-2)$ which is undefined) ①



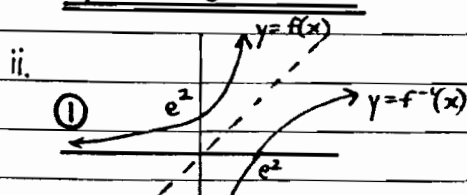
ii. $\underline{x \geq -1}$ ①

Question 2 (8 marks)

a) i. $y = e^{x+2}$
 $x = e^{y+2}$ ①

$\log_e x = y+2$

$\underline{y = \log_e x - 2}$ ①



b) $P(x) = x^3 + 2x^2 + ax + b$

$(x+2)$ is a factor so $P(-2) = 0$ ①

$-8 + 8 - 2a + b = 0$

$-2a + b = 0$ ①

also $P(2) = 12$ ①

$8 + 8 + 2a + b = 12$

$2a + b = -4$ ②

solve simultaneously ① + ②

$2b = -4$

$b = -2$ ①

$2a - 2 = -4$

$a = -1$ ①

$\therefore \underline{a = -1 \text{ and } b = -2}$

Question 3 (9 marks)

a) $\frac{1}{\sqrt{1-x^2}} \sin^{-1} x = \frac{1}{\sqrt{1-x^2} \sin^{-1} x}$ ①

b) $\int_0^{2/5} \frac{dx}{\sqrt{16-25x^2}}$
 $= \int_0^{2/5} \frac{dx}{\sqrt{25\left(\frac{16}{25}-x^2\right)}}$ ①
 $= \frac{1}{5} \int_0^{2/5} \frac{dx}{\sqrt{\frac{16}{25}-x^2}}$
 $= \frac{1}{5} \left[\sin^{-1} \frac{x}{4/5} \right]_0^{2/5}$
 $= \frac{1}{5} \left[\sin^{-1} \frac{5x}{4} \right]_0^{2/5}$ ①

$= \frac{1}{5} \left(\sin^{-1} \frac{1}{2} - \sin^{-1} 0 \right)$
 $= \frac{1}{5} \left(\frac{\pi}{6} - 0 \right)$
 $= \underline{\underline{\frac{\pi}{30}}}$ ①

c) $\int_0^{\sqrt{3}} \frac{e^x}{1+e^{2x}} dx$ $u = e^x$
 $\frac{du}{dx} = e^x$
 $du = e^x dx$
 $= \int_1^{\sqrt{3}} \frac{du}{1+u^2}$ ①

$$\begin{aligned}
 &= [\tan^{-1} u]_1^{\sqrt{3}} \quad (1) \\
 &= \tan^{-1} \sqrt{3} - \tan^{-1} 1 \\
 &= \frac{\pi}{3} - \frac{\pi}{4} \\
 &= \frac{\pi}{12} \quad (1)
 \end{aligned}$$

Question 4 (8 marks)

$$y = \log_e \left(\frac{2x}{2+x} \right) \quad \begin{matrix} x < -2, \\ x > 0 \end{matrix}$$

$$a) 0 = \log_e \left(\frac{2x}{2+x} \right)$$

$$\begin{aligned}
 1 &= \frac{2x}{2+x} \\
 2+x &= 2x \\
 \therefore x &= 2 \quad (1)
 \end{aligned}$$

$$b) y = \log_e 2x - \log_e (2+x)$$

$$\begin{aligned}
 \frac{dy}{dx} &= \frac{2}{2x} - \frac{1}{2+x} \\
 &= \frac{1}{x} - \frac{1}{2+x} \quad (1) \\
 &= \frac{2+x-x}{x(2+x)}
 \end{aligned}$$

$$\therefore \frac{dy}{dx} = \frac{2}{x(2+x)}$$

$\frac{2}{x(2+x)} > 0$ for all x in the given domain

$\therefore$ function is increasing for all x in the given domain as $\frac{dy}{dx} > 0$ (1)

c) Inflexions: $\frac{d^2y}{dx^2} = 0$ & concavity changes

$$0 = \frac{1}{(2+x)^2} - \frac{1}{x^2}$$

$$\frac{1}{x^2} = \frac{1}{(2+x)^2}$$

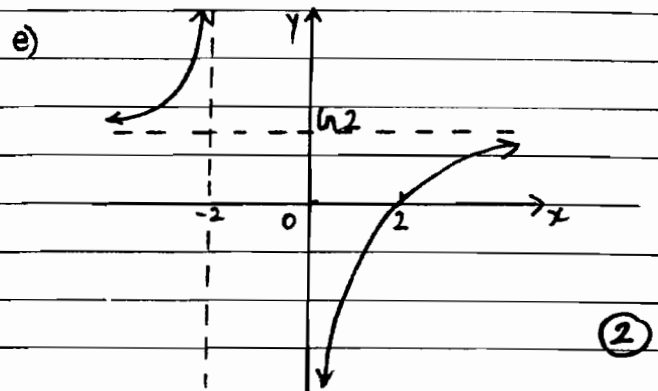
$$x^2 = x^2 + 4x + 4$$

$$x = -1 \quad (\text{out of domain}) \quad (1)$$

$\therefore$ no points of inflexion (1)

$$\begin{aligned}
 d) \lim_{x \rightarrow \infty} \left[\log_e \frac{2x}{2+x} \right] \\
 &= \lim_{x \rightarrow \infty} \left[\log_e \frac{\frac{2x}{x}}{\frac{2}{x} + \frac{x}{x}} \right] \\
 &= \lim_{x \rightarrow \infty} \left[\log_e \frac{2}{\frac{2}{x} + 1} \right] \\
 &= \log_e 2
 \end{aligned}$$

$\therefore$ $y = \log_e 2$ is horizontal asymptote (1)



Question 5 (8 marks)

$$\begin{aligned}
 u &= x & v &= \tan^{-1} x \\
 u' &= 1 & v' &= \frac{1}{1+x^2}
 \end{aligned}$$

a) i.

$$\frac{d}{dx} (x \tan^{-1} x) = \tan^{-1} x + \frac{x}{1+x^2} \quad (1)$$

$$ii. \tan^{-1} x = \frac{d}{dx} (x \tan^{-1} x) - \frac{x}{1+x^2}$$

$$\begin{aligned}
 \int_0^1 \tan^{-1} x \, dx &= \int_0^1 \frac{d}{dx} (x \tan^{-1} x) \, dx \\
 &\quad - \int_0^1 \frac{x}{1+x^2} \, dx \quad (1)
 \end{aligned}$$

$$= \left[x \tan^{-1} x - \frac{1}{2} \ln(1+x^2) \right]_0^1 \quad (1)$$

$$= \left(\tan^{-1} 1 - \frac{1}{2} \ln 2 \right) - 0$$

$$= \frac{\pi}{4} - \frac{1}{2} \ln 2 \quad (1)$$

b) $P(x) = 3x^3 - bx^2 - 27x + 9$
 let the roots be $\alpha, \frac{1}{\alpha}, -\alpha$

iv. $y = \sin^{-1}(x-1)$
 $\sin y = x-1$
 $x = \sin y + 1$

i. sum of roots :

$$\alpha + \frac{1}{\alpha} - \alpha = \frac{b}{3}$$

$$\frac{1}{\alpha} = \frac{b}{3}$$

$$b = \frac{3}{\alpha}$$

①

product of roots :

$$\alpha \cdot \frac{1}{\alpha} \cdot -\alpha = -\frac{9}{3}$$

$$-\alpha = -3$$

$$\alpha = 3$$

$$\therefore \underline{\underline{b = 1}}$$

①

ii. the roots are $3, \frac{1}{3}, -3$ ①

$$\therefore \underline{\underline{P(x) = (x+3)(x-3)(3x-1)}} \quad \text{①}$$

Question 6 (8 marks)

$$f(x) = \sin^{-1}(x-1)$$

i. $f(0) = \sin^{-1}(-1)$
 $= \underline{\underline{-\frac{\pi}{2}}}$

①

ii. $-1 \leq x-1 \leq 1$

$$\therefore \underline{\underline{0 \leq x \leq 2}} \quad \text{①}$$

$$R: \underline{\underline{-\frac{\pi}{2} \leq y \leq \frac{\pi}{2}}} \quad \text{①}$$

$$V = \pi \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} (\sin y + 1)^2 dy \quad \text{①}$$

$$= \pi \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \sin^2 y + 2 \sin y + 1 dy$$

* using $\cos 2A = 1 - 2\sin^2 A$
 $\sin^2 A = \frac{1}{2}(1 - \cos 2A)$

$$= \pi \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{1}{2} - \frac{1}{2} \cos 2y + 2 \sin y + 1 dy \quad \text{①}$$

$$= \pi \left[\frac{y}{2} - \frac{1}{4} \sin 2y - 2 \cos y + y \right]_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \quad \text{①}$$

$$= \pi \left[\left(\frac{\pi}{4} - \frac{1}{4} \sin \pi - 2 \cos \frac{\pi}{2} + \frac{\pi}{2} \right) - \left(-\frac{\pi}{4} - \frac{1}{4} \sin(-\pi) - 2 \cos\left(-\frac{\pi}{2}\right) - \frac{\pi}{2} \right) \right]$$

$$= \pi \left[\frac{3\pi}{2} \right]$$

$$= \underline{\underline{\frac{3\pi^2}{2} \text{ units}^3}} \quad \text{①}$$

iii. $\frac{\pi}{2}$
 $f(x) = \sin^{-1}(x-1)$

①

$$1 \quad \frac{1}{2}$$

$$-\frac{\pi}{2}$$